ARITHMETIC CALCULATOR USING PYTHON TKINTER

AIM

To design and develop a simple arithmetic calculator using Python Tkinter. The application allows the user to enter two numbers and perform addition, subtraction, multiplication, division, and remainder operations through a graphical user interface. It also provides a Clear button to erase the entered values and result.

PROCEDURE

1. Import the Tkinter library required to create the graphical user interface.
2. Create the main application window and set its title, size, and background colour.
3. Create labels and entry boxes for Number 1, Number 2, and Result.
4. Define a function for addition and display the sum in the result field.
5. Define functions for subtraction, multiplication, division, and remainder.
6. Create buttons for ADD, SUB, MUL, DIV, REM, and CLEAR.
7. Connect each arithmetic button to its corresponding function.
8. In each operation, read the two input values, calculate the result, and update the result entry box.
9. Define the Clear function to erase both input values and the result.
10. Run the application using the Tkinter mainloop() function.
11. Test the calculator with sample values and verify the output of each operation.

PROGRAM CODE

The following Python program creates the calculator interface and implements the arithmetic operations.

PROGRAM CODE

```
import tkinter

x = tkinter.Tk()
x.title("ARITHMETIC CALCULATOR")
x.geometry("400x320")
x.config(bg="lightgreen")


def adding():
    num_1 = num1_x.get()
    num_1 = eval(num_1)
    num_2 = num2_x.get()
    num_2 = eval(num_2)

    final = num_1 + num_2
    res_x.delete(0, tkinter.END)
    res_x.insert(0, final)


def subtract():
    num_1 = num1_x.get()
    num_1 = eval(num_1)
    num_2 = num2_x.get()
    num_2 = eval(num_2)

    final = num_1 - num_2
    res_x.delete(0, tkinter.END)
    res_x.insert(0, final)


def multi():
    num_1 = num1_x.get()
    num_1 = eval(num_1)
    num_2 = num2_x.get()
    num_2 = eval(num_2)

    final = num_1 * num_2
    res_x.delete(0, tkinter.END)
    res_x.insert(0, final)


def dividing():
    num_1 = num1_x.get()
    num_1 = eval(num_1)
    num_2 = num2_x.get()
    num_2 = eval(num_2)

    final = num_1 / num_2
```

PROGRAM CODE (continued — 2)

```
    res_x.delete(0, tkinter.END)
    res_x.insert(0, final)


def reminder():
    num_1 = num1_x.get()
    num_1 = eval(num_1)
    num_2 = num2_x.get()
    num_2 = eval(num_2)

    final = num_1 % num_2
    res_x.delete(0, tkinter.END)
    res_x.insert(0, final)


def clearing():
    num1_x.delete(0, tkinter.END)
    num2_x.delete(0, tkinter.END)
    res_x.delete(0, tkinter.END)


g = ("georgia", 15)

num1 = tkinter.Label(
    x, text="Enter Number 1", bg="white",
    fg="black", font=g
)
num1.place(x=20, y=40)

num1_x = tkinter.Entry(
    x, bg="white", fg="black", font=g,
    width=15, cursor="watch"
)
num1_x.place(x=180, y=40)

num2 = tkinter.Label(
    x, text="Enter Number 2", bg="white",
    fg="black", font=g
)
num2.place(x=20, y=90)

num2_x = tkinter.Entry(
    x, bg="white", fg="black", font=g,
    width=15, cursor="watch"
)
num2_x.place(x=180, y=90)

res = tkinter.Label(
```

PROGRAM CODE (continued — 3)

```
    x, text="Result", bg="white",
    fg="black", font=g, width=12
)
res.place(x=20, y=140)

res_x = tkinter.Entry(
    x, bg="white", fg="black", font=g,
    width=15, cursor="watch"
)
res_x.place(x=180, y=140)

add = tkinter.Button(
    x, text="ADD", bg="blue", fg="white",
    font=("georgia", 12), command=adding,
    cursor="tcross"
)
add.place(x=20, y=200)

sub = tkinter.Button(
    x, text="SUB", bg="blue", fg="white",
    font=("georgia", 12), command=subtract,
    cursor="exchange"
)
sub.place(x=80, y=200)

mul = tkinter.Button(
    x, text="MUL", bg="blue", fg="white",
    font=("georgia", 12), command=multi,
    cursor="cross"
)
mul.place(x=140, y=200)

quo = tkinter.Button(
    x, text="DIV", bg="blue", fg="white",
    font=("georgia", 12), command=dividing,
    cursor="clock"
)
quo.place(x=205, y=200)

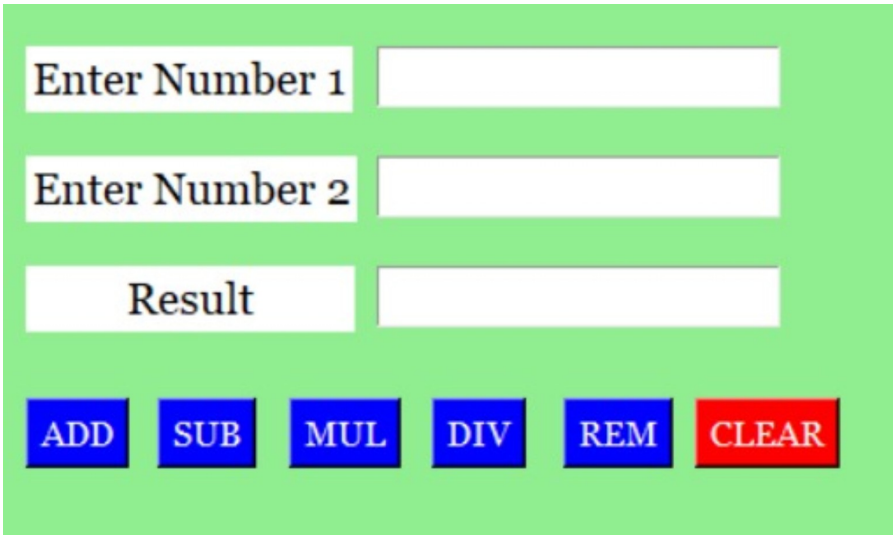
rem = tkinter.Button(
    x, text="REM", bg="blue", fg="white",
    font=("georgia", 12), command=reminder,
    cursor="dotbox"
)
rem.place(x=265, y=200)

clear = tkinter.Button(
    x, text="CLEAR", bg="red", fg="white",
    font=("georgia", 12), command=clearing,
    cursor="heart"
)
clear.place(x=325, y=200)

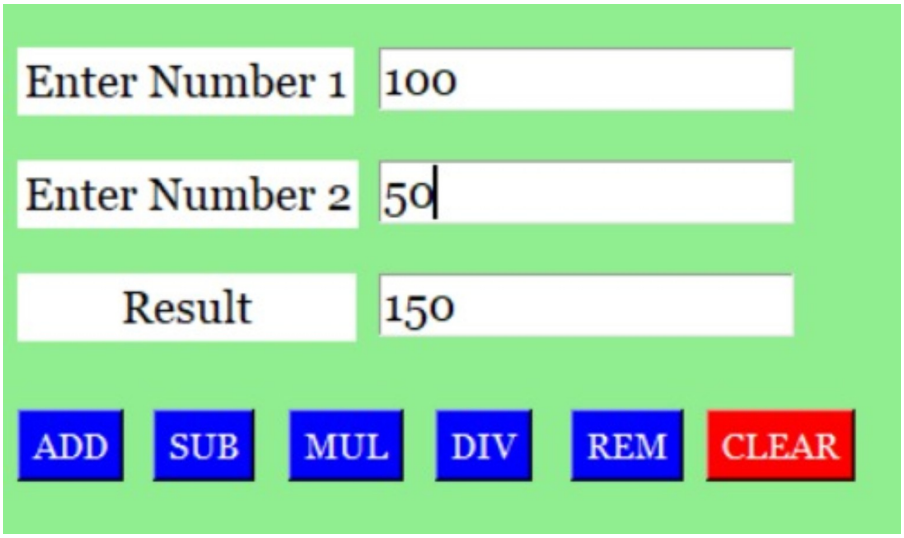
x.mainloop()
```

OUTPUT

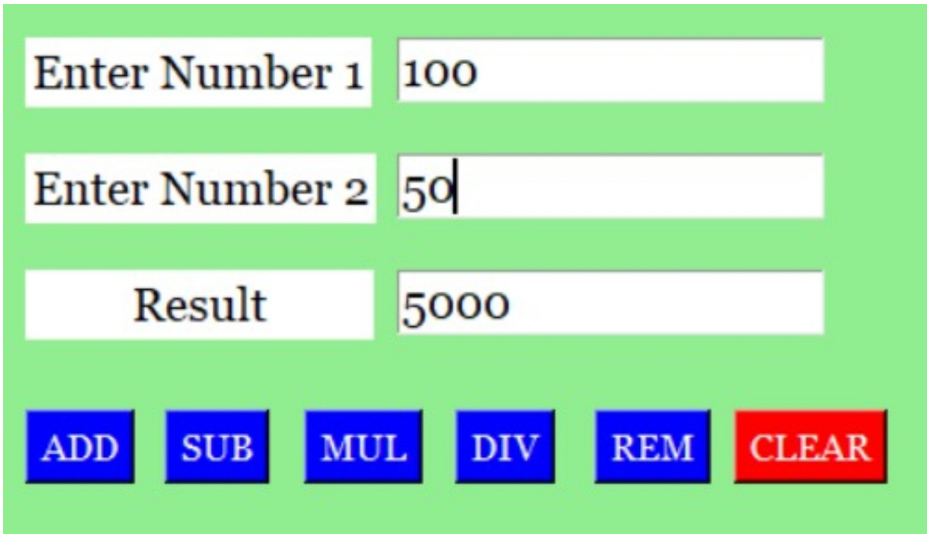
The screenshots below show the calculator interface and sample arithmetic results.



Output 1: Calculator interface before entering values.



Output 2: Addition of 100 and 50 gives 150.



Output 3: Multiplication of 100 and 50 gives 5000.

RESULT

Thus, an arithmetic calculator was developed successfully using Python Tkinter. The program provides a graphical interface with input fields, a result field, and buttons to perform addition, subtraction, multiplication, division, and remainder operations. The Clear button resets the input and result fields. The sample outputs confirm that the calculator can perform arithmetic operations and display the result.